

Epic 4: Model Building

Story 5: Model Comparison and Selection

After training all the machine learning models, a comparison is performed to evaluate their performance and identify the most suitable model for flood prediction. The `compareModel()` function is used to display the evaluation results of all trained models in a single view, making it easier to compare their predictive performance.

Models Evaluated

The following machine learning classification models were trained and evaluated:

- **Decision Tree**
- **Random Forest**
- **K-Nearest Neighbours (KNN)**
- **XGBoost**

Evaluation Metrics

Each model is evaluated using the following standard performance metrics:

- **Accuracy Score** – Measures the percentage of correctly classified instances.
- **Confusion Matrix** – Displays the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- **Classification Report** – Provides detailed metrics including Precision, Recall, F1-Score, and Support for each class.

Model Performance

The performance comparison showed that Decision Tree, Random Forest, and XGBoost achieved an accuracy of 96.55% on the test dataset. Although multiple models achieved the same accuracy, XGBoost was selected as the final deployment model because of its superior generalization capability, stability, and ability to handle complex feature relationships effectively.

Why XGBoost Was Selected

- **Achieved 96.55% prediction accuracy.**
- **Provides excellent generalization on structured datasets.**
- **Uses a boosting technique that improves prediction performance by learning from previous errors.**
- **More robust and less prone to overfitting than a single Decision Tree.**

- Handles complex data patterns efficiently and produces reliable predictions.

Deployment

The trained XGBoost model is saved using Pickle (.pkl) or Joblib. During deployment, the Flask web application loads the saved model and uses it to generate real-time flood risk predictions based on user-provided weather parameters.

Outcome

- Compared the performance of all machine learning models.
- Evaluated each model using Accuracy, Confusion Matrix, and Classification Report.
- Selected XGBoost as the final model for deployment.
- Saved the trained model for integration into the Flask web application.

Figure 16: Comparison of Decision Tree, Random Forest, KNN, and XGBoost models, showing XGBoost as the final selected model with 96.55% accuracy.

```
[38]: from sklearn import metrics

[39]: print(metrics.accuracy_score(y_test,p1))
      print(metrics.accuracy_score(y_test,p2))
      print(metrics.accuracy_score(y_test,p3))
      print(metrics.accuracy_score(y_test,p4))

0.9655172413793104
0.9655172413793104
0.896551724137931
0.9655172413793104
```